

Unit-I (Atomic & mol.)

- 1) Significance of ψ and ψ^2 .
- 2) Schrodinger wave Eqⁿ.
- 3) CFT and MOT.
- 4) π -molecular orbital for Butadiene & Benzene
- 5) Role of doping in metals.

Unit-II (Spectroscopy)

- 1) UV, IR, NMR main points & applications
- 2) Chromophores and auxochromes
- 3) Fluorescence & Phosphorescence
- 4) Intensity shifts and its 4 types.
- 5) Chemical shift, factors affecting
- 6) Beer's Lambert's law
- 7) Diffraction & types of diffraction

Unit-III (Intermolecular forces)

- 1) Intermolecular forces & types of interactions
- 2) Compressibility factor and cases arise for real gas
- 3) Deviations of real gases from gas law & its causes
- 4) van der Waals Eqⁿ
- 5) PES main imp. questions from (Pdf of unit)

Unit IV (Use of free Energy)

- 1) Entropy, Enthalpy, Gibbs free Energy
- 2) Thermodynamic functions derivations.
- 3) Nernst eqⁿ & its Applications.
- 4) Concepts of Acid & Base (diff. Theories ideas).
- 5) Wet & Dry corrosion & types of corrosion
- 6) Sludge formation in water chemistry
- 7) Ellingham diagram & its limitations

Unit V (Periodic Properties)

- 1) Effective Nuclear charge, Slater's rule and Fajon's rule.
- 2) VSEPR theory
- 3) Factors affecting periodic table.
- 4) HSAB principle
- 5) Polarizability & factors influencing.

Unit VI (Stereochemistry)

- 1) Isomerism & its types.
- 2) Stereoisomers and types
- 3) Enantiomers and Diastereomers
- 4) Fischer projection

Unit - VII
(Organic Rxn. & Synthesis of Drug)

- 1) Nucleophilic sub. Rxn., Elimination Rxn., electrophilic Rxn.
- 2) S_N1 , S_N2
- 3) cyclization reaction.
- 4) Synthesis of Paracetamol & Aspirin.